EE2004 Digital Signal Processing

Tutorial 4 Answer Key

1. (a) $|z| > \frac{4}{3}$, $x[n] \notin \ell_1$, (b) $0 < |z| < \frac{10}{9}$, $x[n] \in \ell_1$, (c) $\frac{e}{\pi} < |z| < \infty$, $x[n] \in \ell_1$, (d) $\frac{e}{\pi} < |z| < \sqrt{13}$, $x[n] \in \ell_1$, (e) $|z| > 1$, $x[n] \notin \ell_1$, (f) $|z| > \frac{1}{2}$, $x[n] \in \ell_1$.

2. (a) $G(z) = z^{-(N-1)} H^*(1/z^*)$, (b) $1/z_0^*$ is also a zero, (c) z_0^*, $1/z_0^*$, $1/z_0$.

3. (a) False, (b) False, (c) False (Hint: consider poles at $z_k = r\, e^{j\frac{2\pi k}{N}}$ for $k = 0, 1, \ldots, N-1$).

4. $H(z) = \dfrac{2}{(1 - 0.4\, z^{-1})(1 + 0.2\, z^{-1})}$, RoC: $|z| > 0.4$.

5. $|a| < 1/r$.

6. 4

7. $x[n] = n\, u[n]$ (assuming causality). (a) $z^{-3}\, X(z)$, (b) $3\, z^{-3}$, (c) $(1 - z^{-1})X(z)$, (d) $-z^{-1}$, (e) $5z^{-1}X(z) + \dfrac{4}{1 + \frac{1}{3}z^{-1}}$

8. (a) $x_1[n] = -\frac{a^n}{n}\, u[n-1]$, (b) $x_2[n] = -\frac{a^{-n}}{n}\, u[n-1]$, (c) $x_3[n] = \frac{a^n}{n}\, u[-n-1]$

9. $x[n] = \{\overset{\downarrow}{2}, -2, -1, 5, -6, -2\}$.

10. (a) $X(z) = \dfrac{1}{1 - 0.3\, e^{j0.25\pi} z^{-1}}$ with RoC $|z| > 0.3$, (b) $X^*(1/z^*) = \dfrac{1}{1 - 0.3\, e^{-j0.25\pi} z}$ with RoC $|z| < 10/3$, (c) $R_{xx}(z) = \dfrac{1}{1.09 - 0.3(e^{-j0.25\pi} z + e^{j0.25\pi} z^{-1})}$ with RoC $\frac{3}{10} < |z| < \frac{10}{3}$, (d) $r_{xx}[n] = 1.098 \cdot 0.3^{|n|}\, e^{j0.25\pi n}$